C5 past final worksheet

1. (Fall 2024, #8) We intend to find the area of the region bounded by the graph of $f(x) = \dfrac{1}{1+x}$, the x-axis and the lines $x = 0$ and $x = 0.1$. To that end, we approximate f near $x = 0$ by a Taylor polynomial. What should the degree of the polynomial be at least to achieve an accuracy of 0.01?

2. (Spring 2024, #9) Let $T_3(x)$ denote the Taylor polynomial of degree at most w about $x = \pi/4$ of $f(x) = \tan x$. Notice the first few derivatives of f are:

$$\begin{aligned}
f(x) &= \tan x \\
f'(x) &= \sec^2 x = 1 + \tan^2 x \\
f''(x) &= 2\tan x + 2\tan^3 x \\
f'''(x) &= 6\tan^4 x + 8\tan^2 x + 2 \\
f^{(4)}(x) &= 24\tan^5 x + 40\tan^3 x + 16\tan x
\end{aligned}$$

You do not need to show these.

(a) Find $T_3(x)$.

(b) Find a bound on the error when $T_3(x)$ is used to approximate $\tan(x)$ for $0 \le x \le \pi/4$.

3. (Fall 2023, #9) Let $f(x) = x^2 \ln x$.

   (a) Compute $T_3(x)$ at 1.

   (b) What is the maximum possible error in the approximation $f(x) \sim T_3(x)$, when $1/2 \leq x \leq 3/2$.

4. (Spring 2023, #8) Consider the function $g(x) = (x - 1)\ln x$.

(a) Find the degree-2 Taylor polynomial, $T_2(x)$, of g centered at $a = 1$.

(b) Use $T_2(x)$ to approximate $g(1.5)$.

(c) Estimate the accuracy of the above approximation.

5. (Fall 2021, #8) Let $f(x) = x^2 \ln(x+1)$. Find the 2nd-order Taylor polynomial of $f(x)$ centered at $x = 1$.